# Taeyun Roh

Seoul, Republic of Korea | txxnrd.github.io | github.com/txxnrd

AI Researcher building autonomous agents powered by advanced visual reasoning to thoroughly understand their environments.

## Education

**Korea University** Mar. 2026 – Present
*M.S. in Computer Science and Engineering (GPA: 4.5 / 4.5)* Seoul, Republic of Korea

**Korea University** Mar. 2022 – Feb. 2026
*B.S. in Computer Science and Engineering (GPA: 4.08 / 4.5)* Seoul, Republic of Korea

## Research Experience

**Data Mining & Information Systems Laboratory, Korea University** Mar. 2025 – Feb. 2026
*Undergraduate Researcher, advised by Prof. Jae Woo Kang* Seoul, Republic of Korea

- Research on agentic memory systems for AI agents.

**Network and Mobile System Laboratory, KAIST** Jan. 2025 – Mar. 2025
*Research Intern, advised by Prof. Sung-Ju Lee* Daejeon, Republic of Korea

- Worked on efficient LLM inference, including on-device LLM inference.

**System Software Laboratory, Korea University** Jan. 2024 – Dec. 2024
*Undergraduate Researcher, advised by Prof. Gyeongsik Yang* Seoul, Republic of Korea

- Researched federated learning for healthcare applications under resource heterogeneity.

## Selected Publications

**Distilling Expert-level Planning for Real-world Diabetes Prescribing**
Yein Park, Sihyeon Park, Taeyun Roh, Mujeen Sung, Jaehoon Yun, Sooyeon Lee, Sanghoon Lee, Jaewoo Kang
ICML 2026 Workshop on Structured Data for Health

**CLAG: Adaptive Memory Organization via Agent-Driven Clustering for Small Language Model Agents** [Project Page]
Taeyun Roh, Wonjune Jang, Junha Jung, Jaewoo Kang
ACL 2026 Findings

**Breaking Failure Cascades: Step-Aware Reinforcement Learning for Medical Multimodal Reasoning** [Project Page]
Junha Jung, Minbyul Jeong, Suhyeon Lim, Sungwook Jung, Jaehoon Yun, Taeyun Roh, Mujeen Sung, Jaewoo Kang
EMNLP 2026

**BUZZY: Contrastive Scoring to Mitigate Text-Induced Bias in Multimodal Multiple-Choice QA** [Project Page]
Taeyun Roh, Suhyeong Park, Dongyoung Lee, Wonjune Jang, Junha Jung, Eun-yeong Jo, Jaewoo Kang
Under Review

**Learning How to Remember: Memory Modality Selection for GUI Agents** [Project Page]
Taeyun Roh, Suhyeong Park, Gijin Kim, Wonjune Jang, Jaewoo Kang
Under Review

## Industry & Development Experience

**Samsung Research America** Sep. 2026 – Present
*AI Research Intern* Mountain View, CA

- Research on Multimodal Agent Memory System

**Aigensciences** Mar. 2025 – Aug. 2025
*AI Researcher* Seoul, Republic of Korea

- Built an LLM-based pipeline for automated hypothesis generation in drug discovery.

- Developed a pediatric counseling chatbot designed for KakaoTalk integration.

## Awards

- **1st Place, Korea University Capstone Design** (Prize: KRW 1,800,000) — Developed *NEWSQRAB*, an AI-driven system for automated news-to-short-form video generation using RAG-based LLMs.
- **2nd Place, Korea University NEXT Software Competition** (Prize: KRW 1,000,000) — Recognized as lead AI engineer for the *NEWSQRAB* project.
- **3rd Place, 2024 KIISE Conference (Undergraduate Category)** (Dec. 2024) — Awarded for the paper *"Analysis on Research for Data and Resource Heterogeneity in Split-Federated Learning."*
- **2nd Place, 2023 Korea University Hackathon** (Prize: KRW 1,000,000) — Built Noonchi, a real-time visitor prediction and recommendation service for Han River parks.

## Teaching Experience

**Korea University** Fall 2025
*Teaching Assistant, Machine Learning (COSE 362), instructed by Prof. Jae Woo Kang* Seoul, Republic of Korea

## Academic Service

**Conference Reviewer**
*ICLR 2027, ACL 2026 (Secondary), EMNLP 2026(Secondary), EACL 2026 (Secondary)*

**Workshop Reviewer**
*EIML@ICML 2026, PALM@NeurIPS 2026*

## References

**Prof. Jaewoo Kang**
*Professor, Department of Computer Science and Engineering*
Korea University, South Korea
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**Dr. Minbyul Jeong**
*Principal Researcher Scientist of Large Language Model Team*
Upstage, South Korea
Email: minstar@upstage.ai